Conditional Pavement Crack Data Generation for Selective Data Augmentation Using GANs

Afshin Shahrestani, Ali Faisal, and Suliman Gargoum

***Abstract*—The adoption of Deep Learning techniques for automated crack detection is hindered by the lack of high-quality, balanced training datasets, a common issue resulting in detection bias and overfitting in model training. Data imbalances, where certain crack types predominate, cause detection bias and overfitting. This study addresses these challenges by implementing Conditional Wasserstein Generative Adversarial Networks (C-WGAN-GP) to selectively generate high-quality and diverse synthetic pavement crack images that augment underrepresented crack types. Images from Crack500 and CrackForest datasets were annotated and categorized into transverse, longitudinal, and alligator cracks. The contributions of the paper are (1) implementing an augmentation solution that selectively generates diverse and realistic pavement crack images, (2) solving mode collapse and training instability, common issues in training GAN models (3) and demonstrating that training with the GAN-augmented dataset improved classification metrics by 5% and achieved an overall accuracy of 83.33%. This approach not only mitigates data imbalance in the field, but also enhances the accuracy and capabilities of AI-driven pavement maintenance solutions**

***Index Terms*—Pavement crack classification, data balancing, generative AI, data augmentation.**

I. Introduction

PAVEMENT cracking is a common distress that affects all types of pavements. It poses significant challenges to road infrastructure by compromising the safety, serviceability, and longevity of the pavement service life [1]. Pavement cracking is often a result of structural and environmental factors such as weather conditions, traffic loading, binder aging, and mix design characteristics [2]. These cracks lead to faster wear of the pavement structural integrity by allowing water to penetrate porous surfaces and deteriorating the materials in the underlying layers. High severity cracks can also compromise road safety by causing vehicles to lose traction with the road surface, especially in harsh weather conditions. This risk also extends to both pedestrians and cyclists [3]. As such, proactive management of pavement cracking through early detection and diagnosis is of key importance to preserving the road infrastructure.

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Manual inspection methods have been traditionally used to detect regions affected by cracking. These methods include visual inspections where an inspector conducts site visits to map and assigned severity to cracks [4]. This process can be time-consuming, laborious and requires engineering technical background which introduces unwanted subjectivity in crack assessments [4]. Automatic pavement crack detection methods have emerged to solve many of the challenges associated with manual inspections. Early works in pavement detection utilized image processing techniques, such as using threshold-based methods to detect cracks by exploiting the difference in color intensity between crack and non-crack points [5], [7],or performing discontinuity based algorithms on pavement images to exploit crack edges of different grey levels and trace crack regions using image gradient magnitude information [8], [9]. While these attempts have been useful, they have a few limitations. Threshold-based methods' performance in detecting cracks often deteriorates as variation in lighting, pavement colors and textures occurs [10]. Discontinuity-based algorithms can detect crack edges, they might struggle with accurately segmenting complex crack patterns, and their performance might degrade in the presence of noise or irregular crack shapes [11].

Machine learning methods have been integrated in all fields of engineering, such as material engineering, to improve various aspects of traditional approaches and have gained impressive results [12]. More recent studies in pavement analysis have also utilized machine learning techniques to detect cracks in pavements through texture manipulation [10], different classifiers [11], and random forests [12].Some authors also utilized different techniques to automatically detect pavement cracks using image processing where the approaches include utilizing region-based active contour models [17], unsupervised multi-scale algorithms with statistical evaluation [18], and vision-based slicing methods with thresholding [19]. Machine learning techniques heavily depend on the quality and diversity of the training data. Class imbalance is the inheritable problem in the formation of pavement cracks. Such an issue leads to poor generalization and inaccuracies [20].

Recent advancements in computational power have allowed researchers to leverage deep neural networks (DNNs) to detect pavement cracks [20], [21]. These techniques address some of the challenges faced by other methods since they can automate crack detection processes, reducing subjectivity and significantly speeding up the assessment process [22]. Moreover, deep learning models or DNNs can learn intricate features in crack patterns which lead to accurate detection

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and classification of various crack types, sizes, and shapes [24]. They also can adapt to new types of cracks without the need for manual parameter tuning, through transfer and active learning.

Despite their numerous advantages, the performance of deep learning techniques heavily depends on the quality of the dataset in hand. Imbalanced datasets, which refer to a significant difference in the number of instances from each group or class of inputs, can lead to poor generalization, misleading evaluation metrics, and biased models [25]. Small datasets in general can also cause overfitting and high variance in the model. These issues are especially prevalent in pavement crack classification and detection tasks, as gathering an adequate dataset consisting of high diversity in samples, and balance between the different crack types, is nearly impossible. Pavement cracks form under different conditions with unequal probabilities, which make gathering a balanced dataset costly, time consuming, and cumbersome given that it is not possible to selectively choose what kind of cracks are present in the collected dataset.

To address these issues, data augmentation techniques have been considered. Data augmentation techniques are a powerful technique in the field of computer vision that artificially expand the dataset through various transformations [26] to improve model training conditions. Some augmentation approaches involve generating new instances of cracked pavement images by applying diverse modifications to the original dataset. resulting in synthetic images that are indistinguishable from real images. By augmenting the dataset, the models used for crack detection can be more robust and accurate across a wider range of scenarios, ultimately improving their performance [26]. Deep generative networks (thoroughly reviewed in section II, Related Works) have been previously utilized as augmentation models in pavement crack datasets. However, these models typically suffer from common issues which hinder the augmentation. Generative models can experience instability in their training and develop a bias towards certain images, reducing the variety and quality of output images. Furthermore, most data augmentation techniques proposed for pavement crack datasets lack control in their generated outputs and require lengthy postprocessing filtering to balance the dataset.

In this paper, we propose a Conditional Wasserstein Generative Adversarial Network with Gradient Penalty (C-WGAN-GP) that retains the advantages of traditional deep generative models, such as Deep Convolutional Generative Adversarial Networks (DCGAN), while addressing their limitations. The proposed model enhances training stability, mitigates mode collapse, and allows for targeted generation of specific crack types. This approach enables the creation of diverse, high-quality pavement crack images for dataset balancing without requiring additional post-processing steps, thereby streamlining the augmentation process. By generating realistic synthetic images for underrepresented crack types, C-WGAN-GP effectively fills gaps in training datasets. The method is evaluated through a comprehensive framework, comparing classification metrics after augmenting training data with C-WGAN-GP to a non-augmented baseline, DCGAN, and WGAN-GP. Results demonstrate that the proposed model resolves issues associated with DCGANs, achieves superior metrics in classification tasks, and matches the performance of non-conditional models. Specifically, incorporating C-WGAN-GP synthetic data improved the detection algorithm's accuracy, recall, and precision by an average of 5 percent.

II. Related Work

Manual crack detection is subjective and time consuming. To address those issues, several alternative methods have been introduced, Specifically, with the introduction and rise to popularity of deep learning models, new works have emerged in the field of pavement crack detection. One of the biggest complexities with deep learning models is the availability of training data. Classic image processing techniques are the most basic form of data augmentation applied to images. However, in some areas, such as pavement crack images, such techniques can be problematic, as techniques like rotation or mirroring can lead to fundamental changes in the characteristics of the cracks that do not appear in real life. As such, deep generative methods can be a useful alternative here. Deep generative models can create quality artificial images with high variety that keep these fundamental characteristics.

GANs have been previously utilized to help remedy the lack of balanced and diverse datasets in pavement crack tasks. One of the instances of utilization of DCGANs is in study [27] to generate pavement crack images, solving the insufficient training data. The data was captured by a UAV with the comparison between the original data model and the augmented one showing an increase of more than 10 percent in the reported accuracy.

Pei et al. [28] addressed the low availability of data for pavement crack detection using deep learning models by employing a DCGAN. They chose a sample of labeled crack images, performed several preprocessing steps to improve the detection task. The preprocessed images were then given to a variational autoencoder (VAE), which created the encoded input data of the DCGAN. The new synthetic crack images, which were fed into a Faster R-CNN network. The new data augmented model showed an improvement in classification compared to the nonaugmented base model.

In [29] the authors proposed a deep learning approach with a GAN network as the image generator, PCGAN, and modified version of YOLOv3, a single-shot image object detection algorithm, for the detection and crack counting, called YOLO-MF. YOLO-MF has acceleration and median flow (MF) algorithms. The YOLOv3 model detects cracks, and MF keeps track of them in moving video frames. The cracks are tracked over the video frames but are counted only once so the total number of cracks can be calculated by the end of footage. Their proposed detection method achieved better results than other detection methods, such as Faster R-CNN and SSD, in small and adjacent frames. Also, their acceleration algorithm achieved six times faster processing time compared to traditional models.

Although GAN-based crack detection models have shown improvements, they have also introduced extra computational costs and overhead. Reference [30] proposed a lightweight

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GAN structure for pavement crack detection and classification. They experimented with Squeeze and Expand (SE), multi-scale convolution (MC), and depth-wise separable convolution (DSC), to improve the computational cost for the network. Their results indicated that GAN-based models with SE in their fully connected layer, and MC and DSC in their transpose convolutional layers significantly reduced the number of parameters in the model, speeding up the training and inference process, while maintaining acceptable classification accuracy.

CrackGAN was introduced in [31], a DCGAN network modified for the pavement crack detection and segmentation tasks. This study tries to solve the "All Black" phenomena, where unavailability of accurate ground truth (GT) images, leads to misclassification of pixels in the images as background. CrackGAN is a crack-patch-only (CPO) supervised generative adversarial network, that generates input data from non-accurate GT for a Fully Convolutional Network (FCN)-based pavement cracks segmentation model. The proposed approach was tested on several datasets and achieved competitive and state-of-the-art results.

In [20], the authors tried to solve the problem of insufficient training data by modifying the DCGAN architecture for the pavement domain and use it to generate synthetic training data. This modified model, APC-GAN, changes the architecture to adapt to the specific characteristics of pavement crack images, such as their size or the need for bigger kernel sizes in the convolutional layers. To segment the cracks in the pictures, an attention infused U-Net, named AttuNet was added to the proposed framework. They used the open-source DeepCrack dataset to test their models and achieved better evaluation metrics, such as mIoU, F1 Score, and Recall, compared to traditional data augmentation and segmentation methods.

As evident in the literature review, the use of GAN models in data augmentation has been proposed and implemented before. However, depending on the data and model architecture, training a GAN model can lead to issues such as mode collapse, training instability, and overall reduction in quality of the synthetic images, which, in a practical domain, would lead to poor results in detection or classification tasks. Also, the generated images need categorizing and filtering to be used in instances where there is an inherited class imbalance, which is common in pavement defect data. This paper proposes solving the aforementioned issues by (i) introducing Wasserstein distance, as a solution to training instability and mode collapse, and (ii) using conditionality, as means of focusing on desired outputs, into the GAN architecture. This combination results in C-WGAN-GP, which is an extremely critical contribution to the use of deep learning techniques in pavement defect classification.

III. Methodology

This paper proposes a Conditional WGAN with Gradient Penalty to selectively generate pavement crack images based on the desired crack type. As already indicated, a Deep Convolutional GAN (DCGAN) model, a variation of GANs, is a deep neural network that uses real data to generate lifelike synthetic images [32]. Although DCGAN is a powerful data generation model, it is one of the first iterations of the GAN model and has room for improvement.

A typical GAN structure consists of a discriminator and a generator network. The generator learns to transform latent vectors or random noise into synthetic data that resembles real data. The discriminator is tasked with differentiating the fake data generated and the real data from each other by classifying them. The optimization process of GAN can be described as a *minimax* game. The discriminator aims to minimize the loss function associated with its classification task, while the generator aims to maximize the loss function associated with fooling the discriminator. Essentially, the generator tries to make images that are as close to the real images as possible, while discriminator tries to find the fake data generated at all times. This gives rise to a competitive dynamic between the two networks [33].

As indicated by the name, the C-WGAN-GP proposed in this paper represents a Wasserstein GAN with conditionality feature (denoted by C) and a gradient penalty (denoted by GP). A Wasserstein GANs (WGAN) [34] is an evolved version of the DCGAN whereby the loss function of DCGAN is replaced with Wasserstein distance to help the discriminator and generator better understand the differences between the generated and real images. This improved understanding helps the stabilization of the training process and prevents the occurrence of mode collapse in the network. To employ the Wasserstein loss function, certain conditions need to be enforced on the loss function, as a result, the Gradient Penalty is employed to further improve model stability. In this paper, we modify the WGAN model to take crack labels alongside images as inputs, context is provided during training, making the generation of selective images from specific classes possible. This ensures that a user can set conditions on the type of crack generated in the synthetic output resolving imbalances in their training data. In the following sections, the mathematical notations, architecture, and loss functions are discussed in more detail.

A. Loss Function

A DCGAN typically uses a cross-entropy loss function when training the neural network. Mathematically, the objective of the discriminator can be written as below:

$$maxV_D(D,G) = E_{x\sim P_d(x)}[logD(x)] + E_{z\sim P_d(z)}[log(1-D(G(z)))] \quad (1)$$

In Equation (1), x is the real data from the distribution $P_d(x)$; z is the noise vector generated randomly from the gaussian distribution $P_d(z)$; D and G are the discriminator and the generator networks, consisting of convolutional and deconvolutional networks, respectively. Equation (2) shows the value function for the generator:

$$maxV_{G+}\left(D, G = E_{z\sim P_d(z)}\right) = E_{z\sim P_d(z)}\left[logD\left(G\left(z\right)\right)\right] \quad (2)$$

Since we are operating in field of pavement crack detection, we need to modify the default DCGAN network from [32]. DCGAN uses smaller 64×64 pixel images as the default input. While this size might be suitable for some tasks, pavement crack images need to be much bigger and more detailed. As

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TABLE I
GAN MODEL ADVANTAGES/DISADVANTAGES

Model	Advantages	Disadvantages
DCGAN	Faster training time Less computational overhead More understandable loss function	Training instability Prone to mode collapse Post-processing required No control over image class
WGAN-GP	Training stability High quality augmented images Less prone to mode collapse	Long training time Higher computational overhead No control over image class
C-WGAN-GP	Selective image generation Less prone to mode collapse Training stability High image quality	Long training time High computational overhead

such, the network was modified to input and output 256×256 pixel images.

One of the issues with DCGAN models is their lack of stability and their high dependency on hyperparameter tuning. In a lot of situations, DCGAN models fail to converge, or lead to model (mode) collapse. Model Collapse is a common phenomenon in DCGAN models, when the model fails to produce diverse outputs and collapses into generating a limited set of samples. Instances of this phenomena can be seen in table I. As a result, newer architectures of GAN models try to remedy these problems with Wasserstein GAN being one of them [35].

GANs utilize two probability distributions, one for the generated output of the generator model P_g, and one for the distribution of the real images P_r. Their goal is to make sure these two distributions are close to each other, indicating the output is high-quality and similar to the original images. To measure the distance of these distributions, Kullback–Leibler divergence, Jensen–Shannon divergence, and Wasserstein distance are used. Jensen–Shannon divergence is the loss function used in most GAN models, including the DCGAN. This loss function is more susceptible to issues with gradient during training, model collapse, and training instability. WGANs, on the other hand, make use of the Wasserstein distance, to solve or improve some of these problems. The equation below shows the loss function for the discriminator network in WGAN.

$$\max_{w \in \mathcal{W}} \mathbb{E}_{x \sim \mathrm{P}_r}\left[f_w(x)\right] - \mathbb{E}_{z \sim p(z)}\left[f_w\left(g_\theta(z)\right]\right. \tag{3}$$

In Equation(3), $f_w(x)$ is the discriminator, or critic in WGAN, output for the real data, and $f_w(g_\theta(z))$ is the discriminator output for the generated data by the generator. The discriminator tries to maximize distance, differentiating the real and fake data, while the generator tries to minimize this equation by generating data similar to the real ones. $f_w(x)$ also obeys the Lipschitz constraint. The Lipschitz constraint is a mathematical condition that ensures the continuity and boundedness of a function and is as follows:

$$|f_w(x_1) - f_w(x_2)| \leq L.|x_1 - x_2| \tag{4}$$

where f_w is the discriminator function, x is the input point and L is the is the Lipschitz constant.

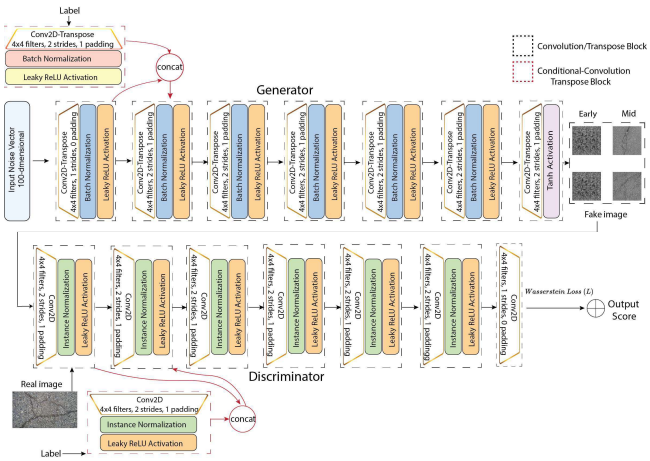

Fig. 1. WGAN-GP and C-WGAN-GP model architectures, consisting of discriminator and generator network. The C-WGAN-GP architecture adds a processing block at the beginning of each network to take labels as additional inputs.

The Lipschitz constraint helps the gradient of the function being bounded, leading to a more stable optimization. To enforce this constraint, weight clipping is usually employed, where weights of the discriminator are bounded within a certain range. This prevents exploding gradient in training. WGANs can take longer to train compared to the normal DCGAN, but the output is often of higher quality and more accurate.

B. Gradient Penalty

Another issue with WGANs is they can still have only poor samples or fail to converge [36] due to weight clipping. To remedy this problem, this paper proposes the use of Gradient Penalty (GP), leading to the creation of WGAN-GP. GP allows for a more stable training network than WGAN and requires very little hyperparameter tuning [36]. The idea behind this method is to add a penalty term to the loss function based on the gradient norm of the discriminator. This penalty is calculated by finding the squared difference of the norm of the gradients and 1. The equation below shows the full loss function with gradient penalty:

$$\begin{aligned} L = {} & \mathbb{E}_{x \sim \mathrm{P}_r}\left[f_w(x)\right] - \mathbb{E}_{z \sim p(z)}\left[f_w\left(g_\theta(z)\right]\right. \\ & + \lambda \mathbb{E}_{\hat{x} \sim \mathbb{P}_{\hat{x}}}[(\|\nabla_{\hat{x}} f_w(\hat{x})\|_2 - 1)^2] \end{aligned} \tag{5}$$

Equation (5) consists of the original WGAN loss plus the gradient penalty term. $P_{\hat{x}}$ is the distribution obtained by uniformly sampling a straight line between the original data distribution and the generated data distribution, P_r and $P_{(z)}$. λ is the penalty coefficient and is used to add weight to the gradient penalty. Another difference between WGAN-GP and DCGAN is the usage of batch normalization. DCGAN makes use of batch normalization to stabilize the training process and speed up the convergence. In WGAN-GP, on the other hand, the gradients of each output with respect to their individual input is desired which is hindered by batch normalization. Figure 1 shows the WGAN-GP model architecture.

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C. Conditionality

Although GANs are great for data generation, there is no control over the generated output. Conditional GANs [37] were introduced to allow conditioning the network with additional information, such as the class of the input, meaning the network can learn the difference between different inputs with the help of the additional provided information. This way, the model can generate specific outputs based on the given information during inference. The additional information can also improve the quality of the generated data.

In the case of pavement cracks, each crack has a specific type and simply creating lines on pavement can lead to the generation of unrealistic crack images. Pavement cracks of different types emerge under certain conditions, and classifying or detecting them accurately can be a good indicator of such characteristics. Having cracks that do not belong to any classes could lead to their misclassification, incorrect results, and faulty decision making.

Our research is focused on four crack types: transverse, longitudinal, and alligator. Each of these crack types have unique and specific qualities differentiating from them. By providing the class label of each crack image alongside the image, the model can better learn to generate different class-specific pavement crack images. To create a conditional WGAN-GP model, the WGAN-GP discriminator and generator architecture needs to be modified, allowing the input of classes, and merging this data with the image data. Figure 1 shows the C-WGAN-GP model architecture.

Conditional WGAN-GP solves the training instability and mode collapse issues occurring in DCGANs by utilizing Wasserstein distance, while allowing for selective conditional pavement generation which improves the post-processing steps normally required by the data augmentation. Table I shows the differences, advantages, and disadvantages of the three mentioned GAN models here.

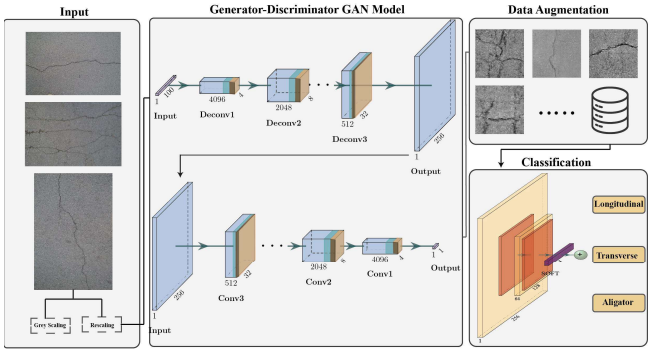

Fig. 2. A high-level overview of the data augmentation pipeline with the data flow.

Fig. 3. Confusion matrix comparison of ResNet18 with and without augmentation with WGAN-GP and C-WGAN-GP.

IV. Implementation and Evaluation Framework

The proposed GAN network is evaluated through a series of classification tasks. A classifier is trained on a selected dataset of alligator, transverse, and longitudinal cracks. DCGAN, WGAN-GP, and C-WGAN-GP models were all trained under the same conditions with the same data to generate pavement crack images from images of the aforementioned three crack types. A baseline for classification was established by training the classifier on the original dataset without any augmentations. The classification results of the datasets augmented with each of these generative networks and baseline are then compared with each other.

A. Data Description

The data used to test the proposed approach consists of publicly accessible datasets, namely, Crack500 [38] and Crack Forest [39]. Crack500 consists of 500 images of size 2000×1500 pixels taken by cell phones on the main campus of Temple University. The images have pixel-level annotations and are separated into train, validation, and test sets. Crack Forest includes 118 pavement crack images of size 480×320 with background noise such as shadows. Since object detection or segmentation were not the goal of this paper, we had to categorize the available images based on the type of crack they featured. The images in the two datasets were combined to create one uniform dataset for pavement crack.

Some of the images from the datasets were removed due to the presence of several crack types in one image, which would cause issues for the classifier. In the end, we had 134 images with alligator cracks, 179 images with longitudinal cracks, and 164 images with transverse cracks. Stratified train test split with a 20% test data split was used to split the dataset into train and test portions. Samples of the original crack images with their classes can be seen in Figure 4.

B. Testing Framework

The testing framework in this paper consists of three modules: *(i) the preprocessing module*, *(ii) the GAN data augmentation module*, and *(iii) the classification module*. Figure 2 is a high-level overview of the data augmentation testing pipeline.

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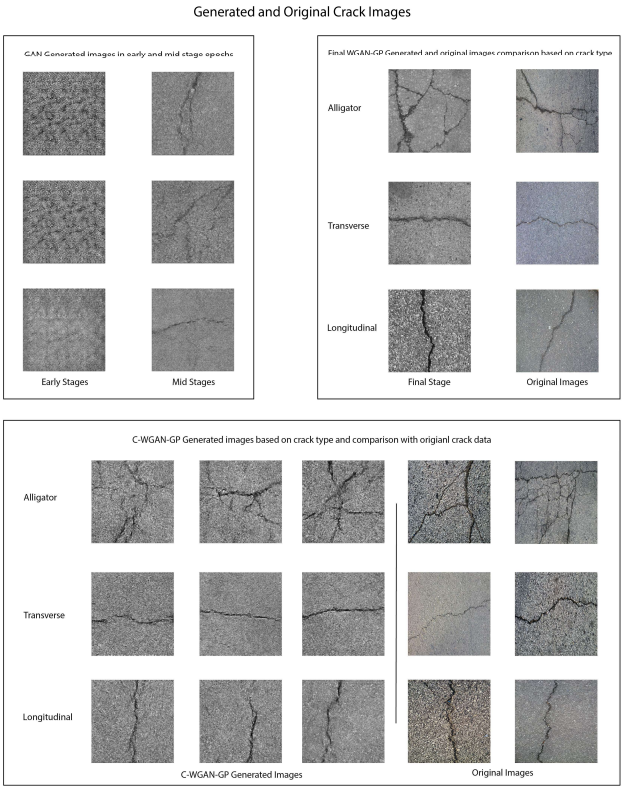

Fig. 4. This figure shows a comparison between synthetic pavement crack images generated by WGAN-GP, C-WGAN-GP, and the original cracks. Both generative models were successful in generating high-quality and diverse samples in 3 distinct crack types after training.

The preprocessing module applies common steps for all GAN models and the final classification model, along with model-specific adjustments. Preprocessing is essential for deep learning models. Images were first converted to grayscale, as early testing showed better classification metrics with 1-channel grayscale images compared to 3-channel RGB images. This is likely due to the nature of pavement images, where black-and-white features dominate, and RGB channels offer no additional benefit. Images were then resized to 256 × 256, normalized, and center-cropped. These steps ensure consistent input shapes and improve training stability by minimizing the risk of exploding gradients. For the GAN module, the aim was to generate synthetic data and augment the real data with those synthetic images to increase the size of the training dataset. We created three different GAN models, DCGAN [32], WGAN-GP [34], and Conditional WGAN-GP, each with their own architecture, loss functions, and hyperparameters. We will discuss the implementation details of each of these models in subsequent sections.

For the classification module, ResNet18 [40] was used as the classifier architecture based on the size of our images and the performance of the model. ResNet18 is one of the implementations of the famous ResNet architecture, with 18 layers to process the images and output the desired results. This network was trained using (real and GAN-generated synthetic images) to categorize the given crack images into alligator, transverse, and longitudinal classes.

1) Model 1: DCGAN: The DCGAN model comprises a discriminator and a generator. The discriminator uses six convolutional blocks to analyze input images and extract features, each including a 2D-convolutional layer (kernel size 4, stride 2, padding 1), batch normalization (except the first block), and leakyReLU activation. This design allows the model to down sample without pooling, improving stability and generalization. The final layer outputs a single value, passed through a sigmoid function for binary cross-entropy loss, which evaluates whether an image is real or generated.

The generator employs reverse architecture, generating artificial pavement crack images from a 100-dimensional random input vector. It consists of transpose convolution blocks, each with a transpose convolution layer, batch normalization, and ReLU activation. These blocks up sample the input to increase spatial resolution. The final layer applies a tanh activation function, converting the input vector into a 256 × 256 grayscale image.

2) Model 2: WGAN-GP: WGAN mostly follows the same architecture for its discriminator and generator as DCGAN, with the difference in the use of instance normalization instead of batch normalization in the discriminator. Instance normalization, as opposed to batch normalization, performs the normalization operation only on the channels and pixels of the instance, not the whole batch. This way, the spatial relations between the pixels are preserved, which is useful for calculating the Wasserstein distance. The convolution blocks in the discriminator use the same hyperparameters as the DCGAN discriminator, begin kernel size of 4, stride of 2, and padding of 1, followed by a convolution layer with kernel size of 4, stride of 1, and padding of 0. Since WGAN-GP does not use the Wasserstein distance instead of the binary cross entropy to calculate the loss of the GAN model, the last layer of the discriminator does not use sigmoid as an activation function and directly outputs the calculated value.

The generator follows the same architecture as DCGAN. Convolution transpose blocks, made up from convolution transpose layers with kernel size of 4, stride of 2 (stride of 1 for the first block), and padding of 1 (padding of 0 for the first block), followed up by batch normalization layers and ReLU activation layers. In the WGAN-GP model itself, during the training, the discriminator or the critic network, is trained more than the generator. In this implementation, the discriminator is first trained for 5 loops and then the generator is trained for 1. During the training of the discriminator, the loss value on real and artificial images is calculated and then gets added to the gradient penalty value. Figure 1 shows the implementation of WGAN-GP with convolution and convolution transpose blocks outlined by black dotted lines.

3) Proposed Model: C-WGAN-GP: To integrate conditionality into the WGAN-GP architecture, modifications are made to both the generator and discriminator. In the generator, the label representing the desired crack type is first passed through a convolutional transpose block, which transforms the single-label input into a spatial feature map. Simultaneously, the input noise vector undergoes a similar transformation. The resulting feature maps from the label and noise are then concatenated into a joint 128×32 matrix that serves as the input to the

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remainder of the generator network. The rest of the generator follows the standard WGAN-GP architecture, consisting of six convolutional transpose blocks and a final tanh activation to produce a 256×256 grayscale image.

The discriminator is similarly modified to accept both an input image and its corresponding label. Each is first passed through a separate convolution block, and their outputs are concatenated into a 128×32 matrix before being forwarded to the remaining convolutional layers of the discriminator. This combined input ensures the discriminator evaluates not just the image realism, but also its consistency with the given label. Figure 1 illustrates the structure of the C-WGAN-GP, highlighting label integration blocks and concatenation points.

The proposed model was evaluated by comparing the accuracy results obtained from the four models discussed in the previous section (baseline, DCGAN, W-GAN, C-WGAN-GP). The ResNet18 classifier was trained for 2000 epochs, with a batch size of 16, learning rate of 0.0002, and beta of 0.5. In each epoch training was monitored on the validation dataset to check accuracy. The training initially consistently increased accuracy but then, due to major similarities between training and validation data, overfitting would occur, resulting in fluctuations in validation metrics.

C. Training and Model Selection

A V100 40GB graphics card from the Digital Alliance CCDB HPC service was used for training GAN models, while an NVIDIA 4080 12GB graphics card handled local image generation and classification tasks. The training process of the classification model was consistently monitored using a fixed validation set across all classification models associated with each GAN. This ensured a controlled environment for proper comparison and evaluation. Early stopping based on validation accuracy was applied, halting training after 200–300 epochs of no improvement. However, this approach occasionally ended training prematurely, so models were also trained fully, with the best instance selected based on validation accuracy, as training loss often proved unreliable due to overfitting.

GAN models were trained on the pavement crack dataset for 10,000 epochs using a learning rate of 0.00005, beta of 0.5, latent space size of 100, and batch size of 32, adjusted for memory availability. Augmentation experiments showed that generating 200 images per class provided optimal improvements, while excessive augmentation increased the risk of overfitting. Generated images were assessed for crack clarity, accurate shapes, and background alignment, though subjective quality alone was not the sole criterion; diversity and feature representation were prioritized.

GAN loss functions primarily facilitate the learning of the discriminator and generator tasks, but they are not reliable indicators of model improvement or output quality. Consequently, regular visual inspection of generated outputs was critical to assess the model's performance. This limitation rendered techniques such as early stopping unsuitable for selecting the optimal generator instance, necessitating manual evaluation to identify models producing high-quality and diverse synthetic images.

TABLE II
AVERAGE CLASSIFICATION METRICS OF RESNET18 WITH/WITHOUT AUGMENTATION

	ResNet18 (Without GAN augmentation)	Model 1* **	Model 2*	Our Proposed Model
Accuracy	79.16	-	**83.33**	**83.33**
Precision	78.04	-	82.46	**82.52**
Recall	77.66	-	82.46	**81.76**
F1-Score	77.62	-	82.46	**81.80**

*Model 1 refers to DCGAN, and model 2 refers to WGAN-GP.
**Training classifier not possible due to imbalanced dataset.

TABLE III
INDIVIDUAL CRACK CLASSIFICATION METRICS OF RESNET18 WITH/WITHOUT AUGMENTATION

	Baseline	**WGAN-GP**	**C-WGAN-GP**
	Alligator		
Precision	69.56	70.37	77.27
Recall	59.25	70.37	62.96
F1-Score	63.99	70.37	69.38
	Longitudinal		
Precision	88.88	86.11	85
Recall	88.88	86.11	94.44
F1-Score	88.88	86.11	89.47
	Transverse		
Precision	75.67	90.90	85.29
Recall	84.84	90.90	87.87
F1-Score	80	90.90	86.56

V. RESULTS AND DISCUSSIONS

C-WGAN-GP was compared to the previously used DCGAN and WGAN-GP architectures to validate the performance and effectiveness of the implemented model under controlled conditions, demonstrating C-WGAN-GP's superiority in classification performance, dataset balance, and augmentation efficiency. The average classification metrics for each trained model can be found in table II, while individual metrics for each class are presented in table III. The classification results for the baseline classifier, trained without any data augmentation, achieved metrics of 78.50 accuracy, 76.32 recall, 77.01 precision, and 76.66 F1-score. These results, presented in Table III, reflect the challenges posed by imbalanced datasets, particularly in accurately identifying complex crack types like alligator cracks. For instance, the precision, recall, and F1-score for alligator cracks were 69.56, 59.25, and 63.99, respectively, showing significant difficulty in differentiating these from other crack types.

The misclassification of alligator cracks can be seen in the model's confusion matrix in Figure 3, which results in the lower recall and precision metrics. This set of metrics serves as the baseline for comparing the effectiveness of the augmentation models.

Model 1, DCGAN, failed to generate images with the necessary diversity and quality to improve baseline performance. It struggled to form complete crack shapes until epoch 7,000,

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with blurring and mode collapse becoming significant beyond epoch 9,000.

Despite producing images of acceptable quality, filtering 5,000 generated samples revealed insufficient variety in crack shapes and backgrounds, rendering the model ineffective for addressing class imbalance.

As a result, retraining the classifier with DCGAN-augmented data was not conducted, as it would have likely introduced further bias and overfitting.

Model 2, WGAN-GP, demonstrated substantial improvements over the baseline, particularly in the classification of alligator cracks. By training the discriminator five times more frequently than the generator, WGAN-GP maintained consistent improvement in output quality until epoch 9,000. The augmented dataset resulted in metrics of 83.33 accuracy, 82.46 recall, 82.46 precision, and 82.46 F1-score, significantly outperforming the baseline. Alligator cracks showed the most notable improvement, within all metrics. Transverse cracks also benefited greatly from the augmentation, achieving a precision, recall, and F1-score of 90.90, compared to the baseline metrics. Although longitudinal cracks showed a slight decrease in precision and F1-score, recall remained consistent at 86.11. These results highlight WGAN-GP's ability to enhance classification performance by addressing imbalances in complex crack types.

Model 3, C-WGAN-GP, incorporating label-specific augmentation, provided comparable overall performance to WGAN-GP while offering additional utility in targeted data generation. The classifier trained on the C-WGAN-GP-augmented dataset achieved metrics of 83.33 accuracy, 81.76 recall, 82.52 precision, and 81.80 F1-score. The model excelled in improving the false positives and false negatives in alligator cracks, owing to the increase in diversity and balance, and achieved much better metrics compared to the baseline. Longitudinal cracks demonstrated the most significant improvement in recall, increasing from 88.88 in the baseline to 94.44, while maintaining competitive F1-scores. Transverse cracks also showed improvement in F1-score compared to the baseline, rising from 80 to 86.56, although slightly lower than WGAN-GP's 90.90.

Both WGAN-GP and C-WGAN-GP effectively addressed the imbalances in the dataset and outperformed the baseline classifier. Alligator cracks, being the most complex, saw the most significant improvements, followed by longitudinal and transverse cracks. While WGAN-GP excelled in recall, particularly for transverse cracks, C-WGAN-GP demonstrated superior precision and offered the added benefit of eliminating the need for post-processing, streamlining the augmentation workflow. This difference stems from the fact that WGAN-GP learns one global crack distribution, so it generates very "archetypal" transverse cracks that the classifier rarely misses (high recall). By contrast, C-WGAN-GP's conditional setup models each crack type separately, producing a wider variety of subtler transverse-crack examples that reduce over-triggering on non-transverse patterns (higher precision).These results, detailed in Table III, underscore the efficacy of the proposed augmentation models in generating high-quality, diverse synthetic images that enhance classifier performance.

A. Custom Training Data Generation

As mentioned in the testing framework section of this paper, to augment the original dataset with WGAN-GP, 10,000 pavement crack images were created and then manually filtered through by a trained expert. To create these images, they had to be created in batches of 2000 images, due to the computational resources the process required and would at least take 15 minutes per batch. This number is dependent on the specifications of the computational system, model size and complexity, and image size. Since each crack type had a different generation probability, to augment a specific crack type with a lower chance of appearance, one might have to filter through thousands of images to get the desired dataset of diverse synthetic cracks. Filtering through 10000 generated images took between one and half to two and half hours. This filtering burden becomes significantly worse for rarer crack types, as the generation in non-conditional models follows the original dataset's class distribution—making it increasingly time-consuming to collect enough samples of infrequent cracks. Although this filtering led to more categorized synthetic pavement crack images than required for the purpose of experimenting with the augmentation effects, C-WGAN-GP managed to generate pre-categorized pavement crack images of the same amount in under 30 minutes. This difference in generation and processing time would further increase if the non-conditional model is used in augmenting a crack type with a lower chance of appearance.

B. Quality of Images

The focused data generation in C-WGAN-GP was rigorously tested with various generated images from the three different available crack types. C-WGAN-GP was able to successfully create the targeted crack types with only minor incorrect instances. Unlike DCGAN, both WGAN-GP and C-WGAN-GP avoided mode collapse and training instability, showing consistent improvement until convergence. The generated images exhibited strong intra- and inter-class diversity and closely resembled real pavement cracks in both shape and characteristics. This visual similarity enabled seamless augmentation without introducing model bias. To ensure fairness, only real pavement crack images were used in the testing set during classification.

Although this study utilized Crack500 and CrackForest due to their common use in benchmarking, the proposed method is expected to scale well to larger and more diverse datasets. The augmentation and detection processes are decoupled: once the synthetic data is generated, the training and deployment of crack detection models proceed as usual. This ensures that the proposed framework remains scalable and practical for real-world applications involving large-scale pavement inspection datasets. Increased variation in crack appearances would likely improve the GAN's ability to learn richer patterns, generate more distinct synthetic samples, and further enhance dataset balance.

Alligator cracks generated by WGAN-GP showed slightly better visual quality, but C-WGAN-GP offered greater shape diversity across all types. Figure 4 shows samples generated by each of these models.

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VI. Limitations

While C-WGAN-GP demonstrated effectiveness in addressing dataset imbalance and improving classification performance for pavement crack detection, it has certain limitations. Training the model is computationally intensive and time-consuming due to the gradient penalty and extended discriminator training, which can pose challenges for those without access to high-performance computing resources. Additionally, the generated image size is constrained by the memory capacity of the hardware, and resizing images to fit these constraints may lead to information loss, though no significant impact was observed in this study. In the evaluation of WGAN-GP, a manual filtering process was used to select class-specific synthetic images, which introduces a degree of subjectivity—an issue C-WGAN-GP was specifically designed to eliminate. Despite these limitations, the method effectively addresses key challenges in data augmentation for pavement crack analysis, significantly enhancing dataset balance and usability.

VII. Conclusion and Future Work

Deep learning technology has the potential to improve the efficiency of pavement surface management. However, the lack of diverse and balanced training datasets remains a major obstacle to fully realizing its potential. In this paper, we introduced Conditional Wasserstein Generative Adversarial Network with Gradient Penalty (C-WGAN-GP), a novel data augmentation method designed to address this challenge. By solving training instability and mode collapse issues while enabling class-focused data generation, C-WGAN-GP offers an efficient and practical solution for dataset balancing. Rigorous testing on benchmark datasets demonstrated the superiority of C-WGAN-GP over DCGAN and its comparability to WGAN-GP, with the added advantage of streamlined data generation processes. Although the method has limitations, such as high computational costs and constraints on image size, these are increasingly manageable with advancements in computing infrastructure. The proposed model offers a valuable tool in enhancing pavement analysis, by making this process cheaper, faster, and more accurate.

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